Data Duplication Removal from Dataset Using Python

Identify Duplicate Rows Using duplicated()

```
# Import pandas
import pandas as pd

# Read the CSV file
df = pd.read_csv("students_duplicates_dataset.csv")

# Display the dataset
print("Original Dataset:\n")
print(df)

# Check for duplicate rows
duplicates = df.duplicated(subset=['Name', 'Age', 'Department', 'City'])

print("\nDuplicate Rows:")
print(duplicates)
```

```
Original Dataset:

   Student_ID     Name  Age Department       City
0         101    Alice   20        CSE    Chennai
1         102      Bob   21      AI&DS  Bangalore
2         103    Alice   20        CSE    Chennai
3         104  Charlie   22        ECE  Hyderabad
4         105      Bob   21      AI&DS  Bangalore
5         106    David   23         IT       Pune
6         107      Eva   20        CSE    Chennai
7         108    David   23         IT       Pune

Duplicate Rows:
0    False
1    False
2     True
3    False
4     True
5    False
6    False
7     True
dtype: bool
```

Remove Duplicates Based on Specific Columns

```
# Import pandas
import pandas as pd

# Read the CSV file
df = pd.read_csv("students_duplicates_dataset.csv")

# Remove duplicates based on selected columns
df_no_duplicates = df.drop_duplicates(
    subset=['Name', 'Age', 'Department', 'City']
)

print("Dataset After Removing Duplicates:\n")
print(df_no_duplicates)
```

```
Dataset After Removing Duplicates:

   Student_ID     Name  Age Department       City
0         101    Alice   20        CSE    Chennai
1         102      Bob   21      AI&DS  Bangalore
3         104  Charlie   22        ECE  Hyderabad
5         106    David   23         IT       Pune
6         107      Eva   20        CSE    Chennai
```

Keep the Last Occurrence of Duplicate Rows

```
# Import pandas
import pandas as pd
```

```
# Read the CSV file
df = pd.read_csv("students_duplicates_dataset.csv")

# Keep the last occurrence of duplicates
df_keep_last = df.drop_duplicates(
    subset=['Name', 'Age', 'Department', 'City'],
    keep='last'
)

print("Keeping the Last Occurrence:\n")
print(df_keep_last)
```

```
Keeping the Last Occurrence:

   Student_ID     Name  Age Department       City
2         103    Alice   20        CSE    Chennai
3         104  Charlie   22        ECE  Hyderabad
4         105      Bob   21      AI&DS  Bangalore
6         107      Eva   20        CSE    Chennai
7         108    David   23         IT       Pune
```

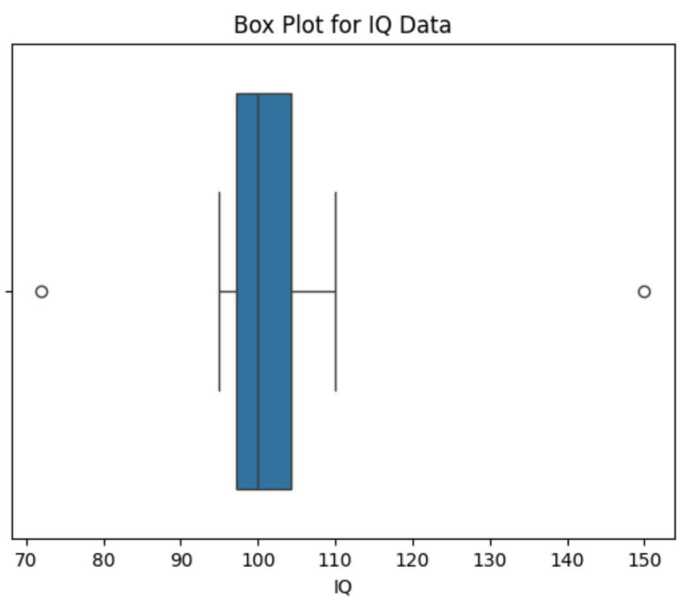
Outliers in Data

Box Plot (Identify Outliers)

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.read_csv("iq_dataset.csv")

sns.boxplot(x=df["IQ"])
plt.title("Box Plot for IQ Data")
plt.show()
```

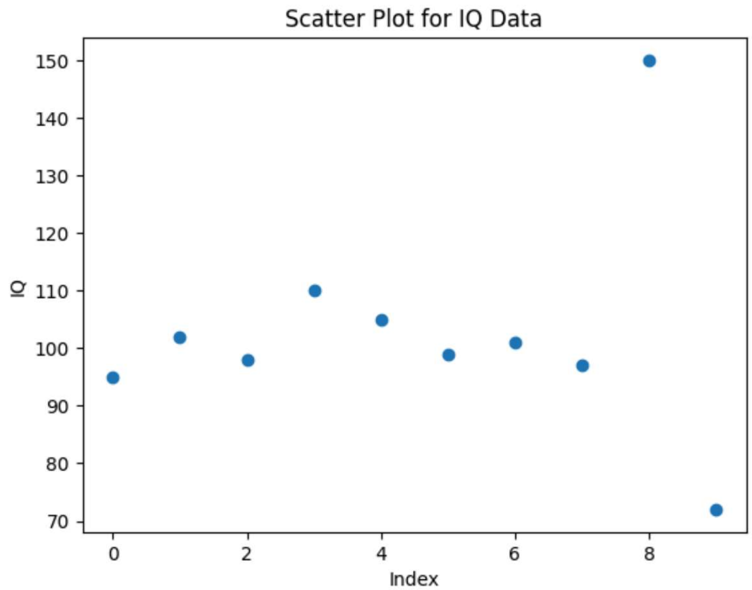


Scatter Plot

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("iq_dataset.csv")

plt.scatter(df.index, df["IQ"])
plt.xlabel("Index")
plt.ylabel("IQ")
plt.title("Scatter Plot for IQ Data")
plt.show()
```



Z-Score Method

```
import pandas as pd
from scipy.stats import zscore

df = pd.read_csv("iq_dataset.csv")

df["Z_Score"] = zscore(df["IQ"])

outliers = df[df["Z_Score"].abs() > 2]

print("Outliers using Z-Score")
print(outliers)
```

```
Outliers using Z-Score
  Student   IQ   Z_Score
8      S9  150  2.566119
```

IQR Method

```
import pandas as pd

df = pd.read_csv("iq_dataset.csv")

Q1 = df["IQ"].quantile(0.25)
Q3 = df["IQ"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

outliers = df[(df["IQ"] < lower) | (df["IQ"] > upper)]

print("Outliers using IQR")
print(outliers)
```

```
Outliers using IQR
  Student   IQ
8      S9  150
9     S10   72
```

FUZZY MATCHING

Read the CSV File

```
import pandas as pd

# Read the CSV file
df = pd.read_csv("customer_data.csv")

print("Customer Dataset")
print(df)
```

Install the Fuzzy Matching Library

```
!pip install thefuzz[speedup]
```

Compare Two Names Using fuzz.ratio()

```
from thefuzz import fuzz

name1 = "John Smith"
name2 = "Jon Smith"

score = fuzz.ratio(name1, name2)

print("Similarity Score:", score)
```

Find the Best Match Using extractOne()

```
from thefuzz import process

query = "John Smyth"

best_match = process.extractOne(query, df["Customer_Name"])

print("Best Match:")
print(best_match)
```

Find Top 5 Matches

```
from thefuzz import process

query = "Alic Jonson"

matches = process.extract(query, df["Customer_Name"], limit=5)

print("Top 5 Matches")
print(matches)
```

Compare Every Name with a Query

```
from thefuzz import fuzz

query = "Michael Lee"

print("Similarity Scores:\n")

for name in df["Customer_Name"]:
    score = fuzz.ratio(query, name)
    print(f"{query}  <-->  {name} = {score}")
```